
EE/CprE/SE 491 WEEKLY REPORT 1

March 5 2025 - March 1 2025

Group number: 16

Project title: Outdoor Roomba

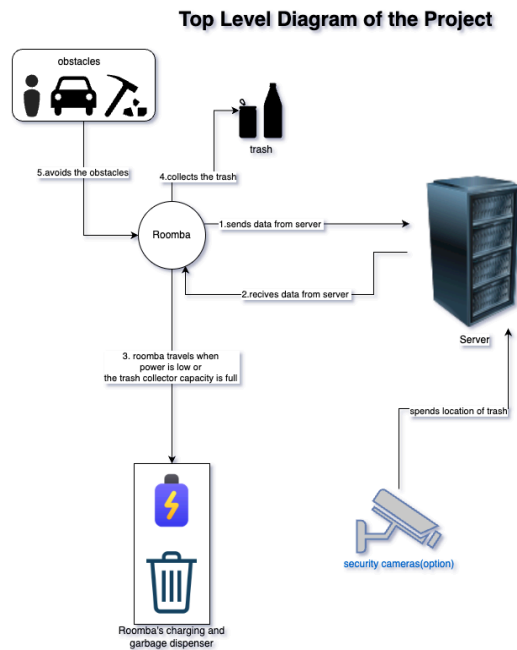
Client &/Advisor: Prof. Diane Rover

Team Members/Role: Brodie M Bates, Daniel Vergara Pinilla ,Mitchell D Owen,Om Shukla, Sudipta Halder

- **Weekly Summary**

Om and Denial work on the Server getting a basic Spring boot working and try to run it on our Ubuntu server from Etg. Brodie has started working on the sweeping device. Mitchel did some refinement on the cad prototype modeling. We have started looking into parts, and discovered prices of various components and have been going up. Therefore we have decided to buy some of the components early, especially the metal and electronics. We made a change to our working prototype design for this semester where instead of using wood we will thin metal sheet similar to our final design. Since metal sheets aren't too expensive, we will properly prototype the early version of chaisses with 3d printers until we think it's good for a metal body. Lastly, our advisor has advised us to find better time to meet up with our team since we will be requiring a lot of brainstorming and design choices in the future.

○ Past week accomplishments



- Sudipta Halder: Looked at metal sheets and prices decided to use them instead of using wood. And Look at a few. Looked some ML models for classification
- Brodie Bates: Designed a few mechanisms for our sweeping device. CAD modeled parts for the mechanism and added parts onto our prototype. Researched different ways we could extend out such as pneumatics and mechanical designs.
- Daniel Pinilla: This week worked on optimization models for the ML device, did research on reducing latency and input lag from the camera module as well as finishing the Database accessibility
- Om Shukla: Learned the Webots software to model the functionality of the robot. Researched more on server related communication between the server and the robot.
- Mitchell Owen: This week was the last week for research and now moving towards 3D modeling. Currently starting to take the base 3D model that Sudipta made and add electrical compartments. Starting to work on the wheels and the Track.

○ Pending issues

- Sudipta Halder: we need a way on discord to track tasks similar to apps like trello or github issues. Still and also work on still finding a better way to meet a lot of members at the same time.
- Brodie Bates: We need more team meetings to fully flesh out the idea and how components need to be setup to have a functioning prototype.

- Daniel Pinilla: Design a code to perform the required subtraction of the trash once it has been detected and cataloged
- Om Shukla: Learn more about Webots and model robot functionality like moving and object detection and object marking in a geofence.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Researched designs for extending out, CAD modeled parts	4	16
Om Shukla	Researched webots and server communications.	5	13
Sudipta Halder	Researched on some parts, And brainstormed some design decisions	2	15
Daniel Mauricio Vergara	Optimization of ML algorithm and reduction of latency of camera model, as well as improvements on db endpoints	3	21
Mitchell Owen	Research and 3D Modeling	4	18

○ **Plans for the upcoming week**

- Sudipta Halder: Will work on making brainstorming meeting. Work on Webots simulation with a few of my team members. Start working on the ML model.
- Brodie Bates: Work on Webots with Sudipta, CAD model more parts and try to simulate a working mechanism to pick up objects. More research into compacting.
- Daniel Pinilla: Design a basic code for socket communication with the db and the raspberry pi
- Om Shukla: Learn more about Webots and model robot functionality like moving and object detection and object marking in a geofence.
- Mitchell Owen: Continuing 3D model and deciding what components to put in the robot and making sure everything fits well. Would like to finish the wheels and start to 3D model the front door and make it move.

○ **Summary of weekly advisor meeting**

In the meeting, She asked us to brainstorm a way to meet more often. Go over requirements a bit more and refine it. Ask us to find better find better time to meet up with our team since we will be requiring a lot of brainstorming and design choices.